



Backend Development in Python

Page No.1.....

Phase 0: Backend Foundation

① What is Backend?

think of Zomato / Amazon:

- Frontend = The waiter / menu you see
- Backend = kitchen
 - Takes your order
 - Check Database
 - Process Payment
 - Store order
 - Send response back

In tech terms:

Backend:

- Receive request (from Frontend)
- Process logic
- Talk to Database
- Return response (Usually JSON)

② How the web works?

client-server model

- client = Browser / APP
- server = Backend app (Python API)

Flow:

Client $\rightarrow$ Request $\rightarrow$ Server

Client $\leftarrow$ Response $\leftarrow$ Server

Example:

① Browser Send HTTP request

: "Give me home page"

② server respond with data



③ What is HTTP? (language of web)

HTTP = Rules for communication between client and server.

Common methods:

- Get → Get
- Post → send data
- Put → update data
- Delete → Delete data

Status code:

- 200 → ok
- 201 → Created
- 400 → Bad request
- 401 → Unauthorized
- 404 → Not Found
- 500 → Server error

Analogy:

- you (client) = Customer
- Waiter (HTTP) = Takes order & send Food
- Kitchen (server) = Prepare response

④ Tiny Backend in Python

From Fastapi import FastAPI

app = FastAPI()

@app.get("/")

def home():

return {"message": "Hello World"}



When you open this in browser:

- ⊙ Browser send Get request
- ⊙ Server return JSON response

Why does the server return JSON (mostly)?

JSON = JavaScript Object Notation

Example:

```
{  
  "name": "Shubham",  
  "role": "Backend Developer",  
  "skills": ["Python", "FastAPI"]  
}
```

it's:

- ⊙ Text-based
- ⊙ Lightweight
- ⊙ Easy for human to read
- ⊙ Supported by almost every language.

Analogy

Think of JSON like a Standard Food Container

- ⊙ You can send any Food inside it
- ⊙ Everyone (browser, mobile app, another server) knows how to open it
- ⊙ It's cheap, light and ~~str~~ standardized.

So companies said:

Let's agree on one simple, universal Format to exchange data".

that Format become JSON.



Do server return only JSON?

No. server can return:

- ⊙ HTML → For website
- ⊙ JSON → For API
- ⊙ Images → .png, .jpg
- ⊙ Files → .pdf, .zip
- ⊙ Plain text → .txt

In FastAPI, Why does it return JSON by default?

Because FastAPI is:

- ⊙ An API Framework
- ⊙ Made for data Exchange
- ⊙ Not for rendering websites

```
@app.get("/")
```

```
def home():
```

```
    return {"message": "Hello"}
```

FastAPI:

- ⊙ Take Python dict
- ⊙ Convert it to JSON
- ⊙ Send it as HTTP response